

Analyzing the Evolution of Research in Economics by Male and Female Economists

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Abstract

Economic research has undergone significant transformations over the past century, driven by evolving paradigms of the state, market, and civil society. Despite increasing attention to diversity in academia, the gendered nature of economic scholarship remains underexplored, with existing studies focusing on short time periods or narrowly defined topics rather than broader paradigmatic themes. This study addresses this gap by conducting a comprehensive, data-driven analysis of the evolution of economic research through a gendered lens. Using Natural Language Processing (NLP) and Deep Learning (DL) techniques, this research examines how topics and themes in economic research have evolved over the past century, distinguishing contributions between male and female economists. The study builds upon the work of Bowles et al. (2025) [1], analyzing publications from major economics journals since 1900. Gender identification is accomplished through a multi-stage process utilizing the World Gender Name Dictionary (WGND 1.0) [5] and Llama for names not found in the dictionary. Topic modeling is conducted using both Latent Dirichlet Allocation (LDA) [2] and BERTopic, a deep learning-based approach [4], leveraging BERT embeddings for nuanced theme detection. Preliminary results suggest notable differences in research focus areas, with female authors tending to emphasize market and civil society topics, while male authors have historically concentrated more on state-related topics. This study enhances understanding of gender diversity in economic scholarship.

Objectives

- To track the evolution of research topics in economics over the past century** and determine whether these trends differ between male and female economists.
- To identify distinct thematic preferences and specialization patterns** based on gender, examining how male and female economists have contributed differently to various economic domains (state, market, and civil society).
- To provide data-driven insights** into gender diversity in economic scholarship using advanced Natural Language Processing and Deep Learning techniques.
- To contribute methodologically** by demonstrating the application of modern computational methods (BERTopic, BERT embeddings) in demographic-aware analysis of academic discourse.

By achieving these objectives, this research seeks to reveal long-term patterns of authorship, thematic focus, and research priorities across genders, identifying areas where gaps persist.

Dataset

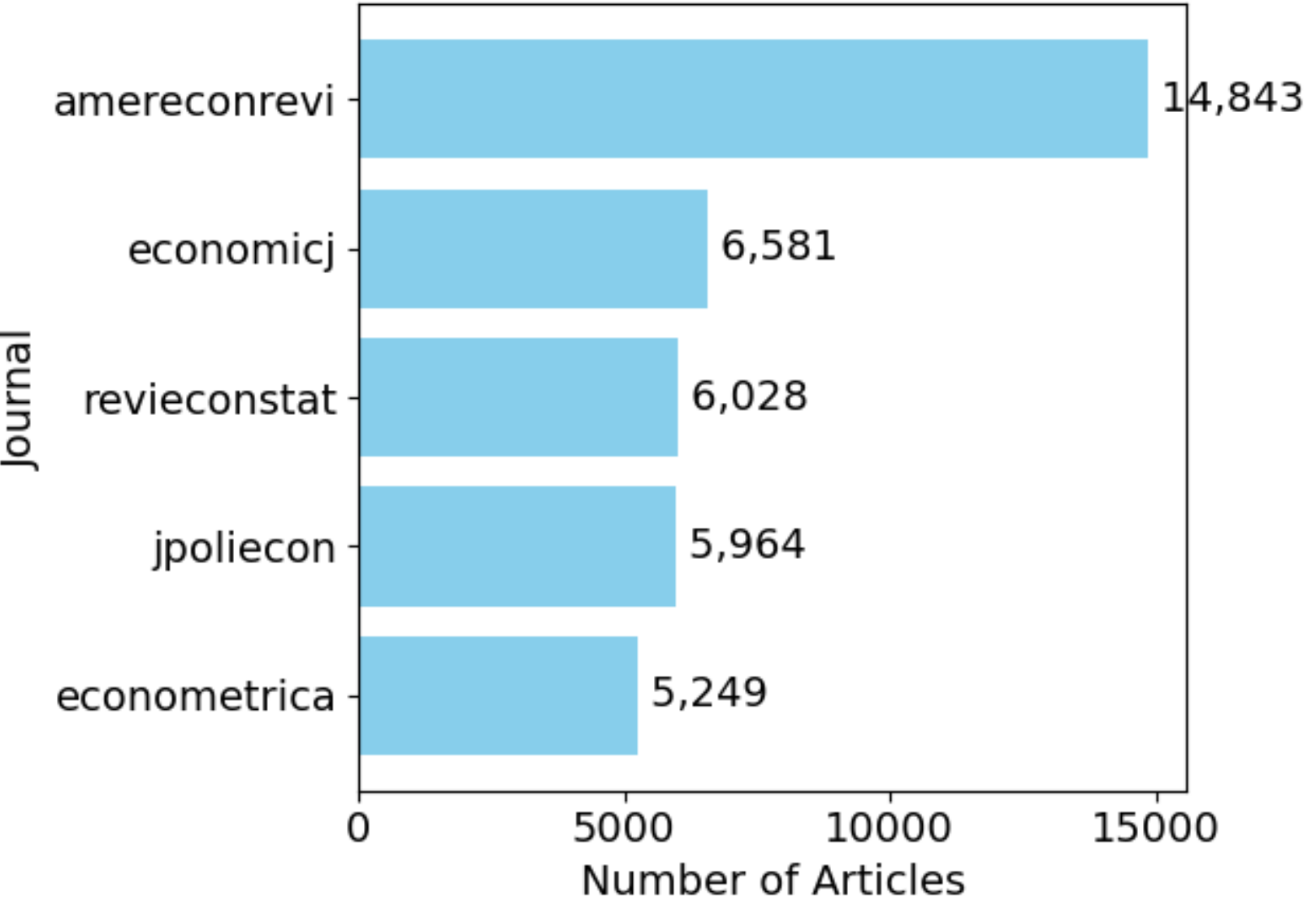
This study uses a comprehensive dataset of economic research papers spanning over a century. The dataset encompasses publications from five major economics journals since 1900, providing a robust foundation for longitudinal analysis of economic scholarship.

Dataset Components:

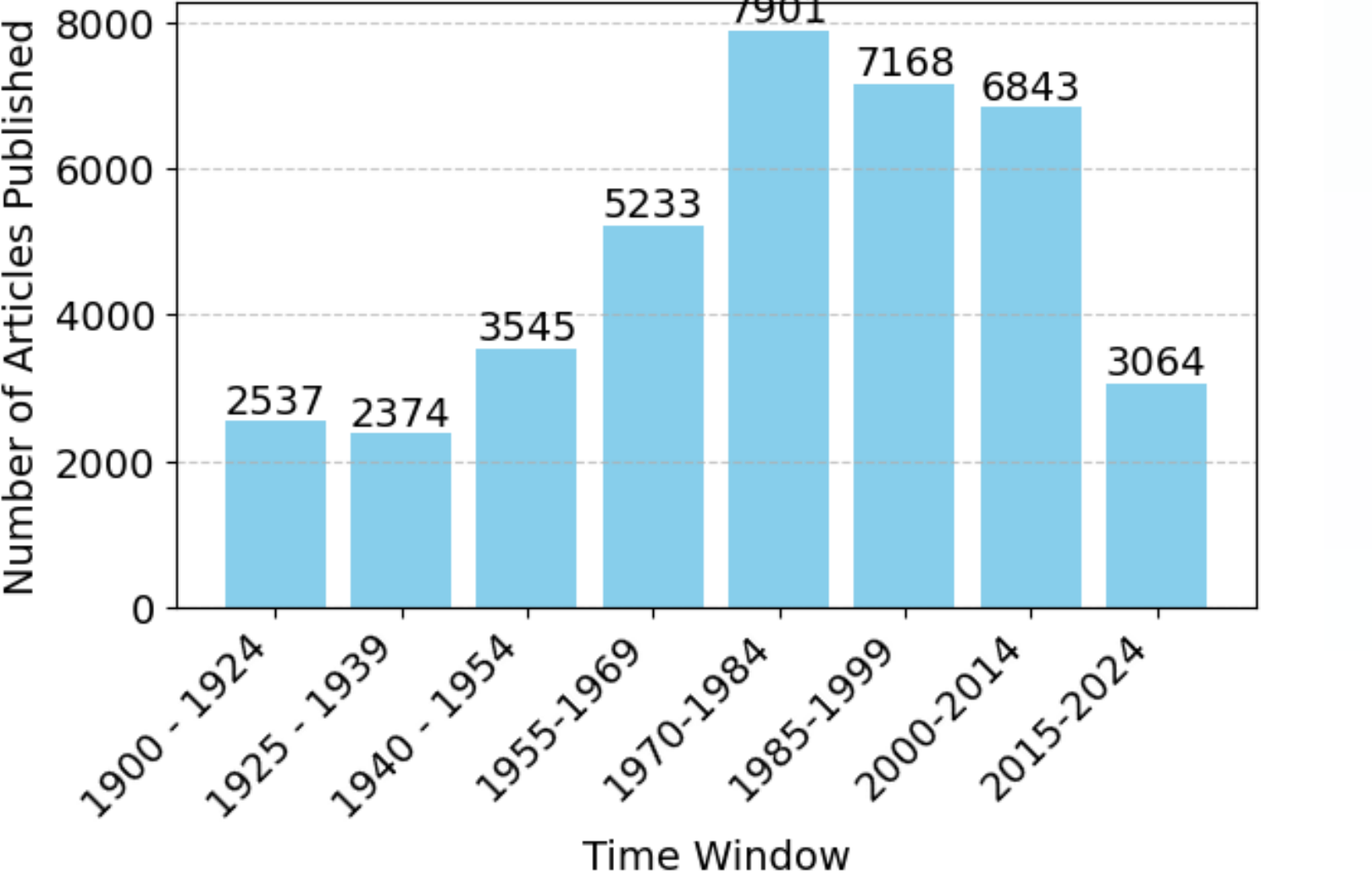
The dataset includes extensive metadata fields extracted from each publication:

- Author names**
- Publication year**
- Full Articles**

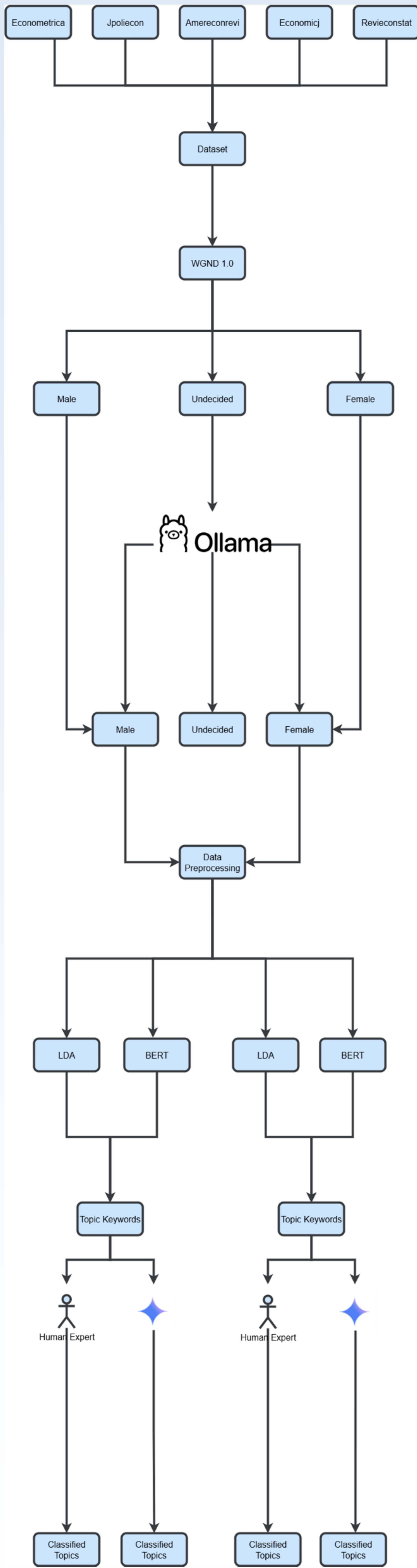
Number of Articles by Journal (Total = 38665)



Number of Articles Published Over Time (Total: 38,665)



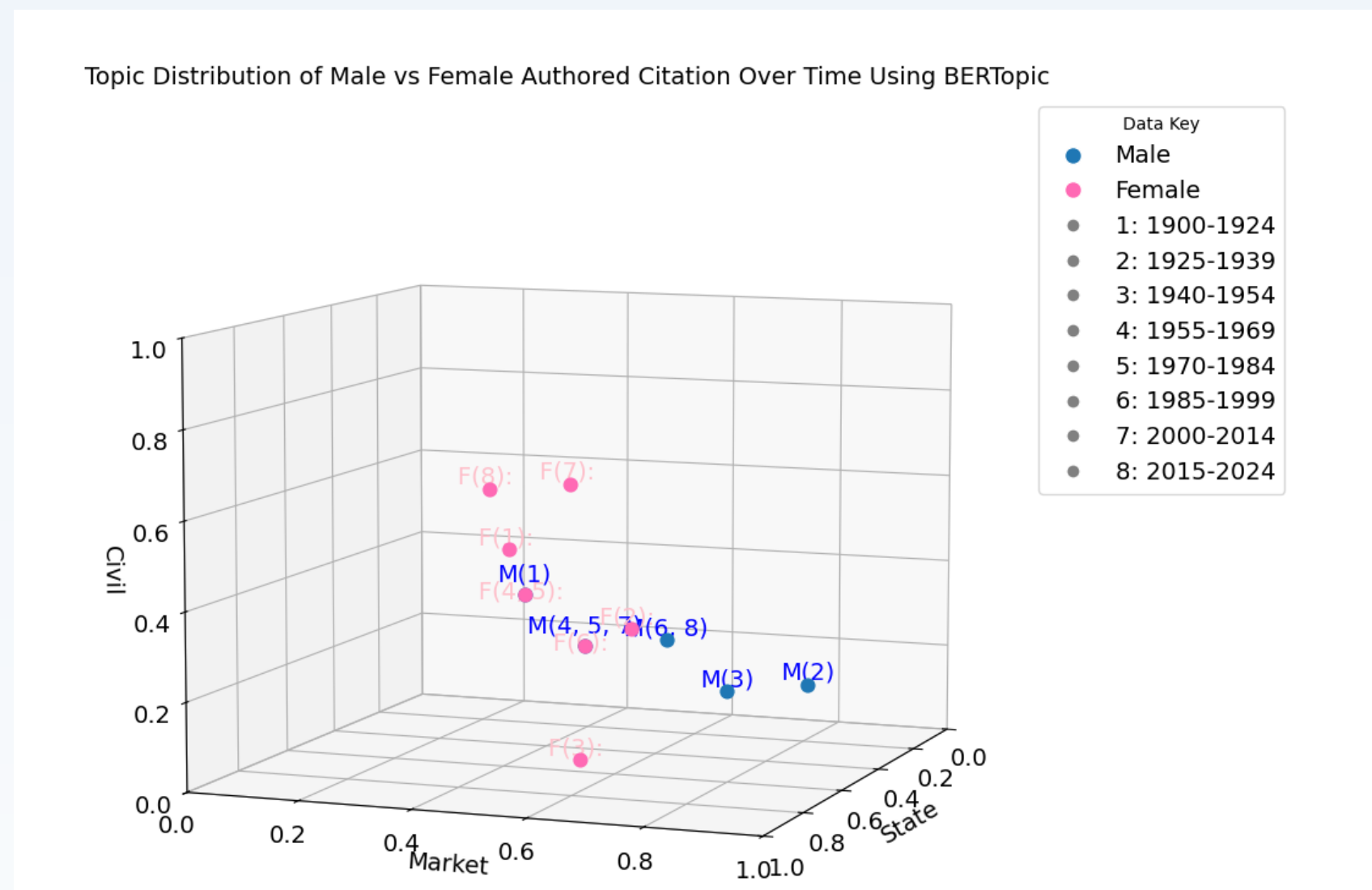
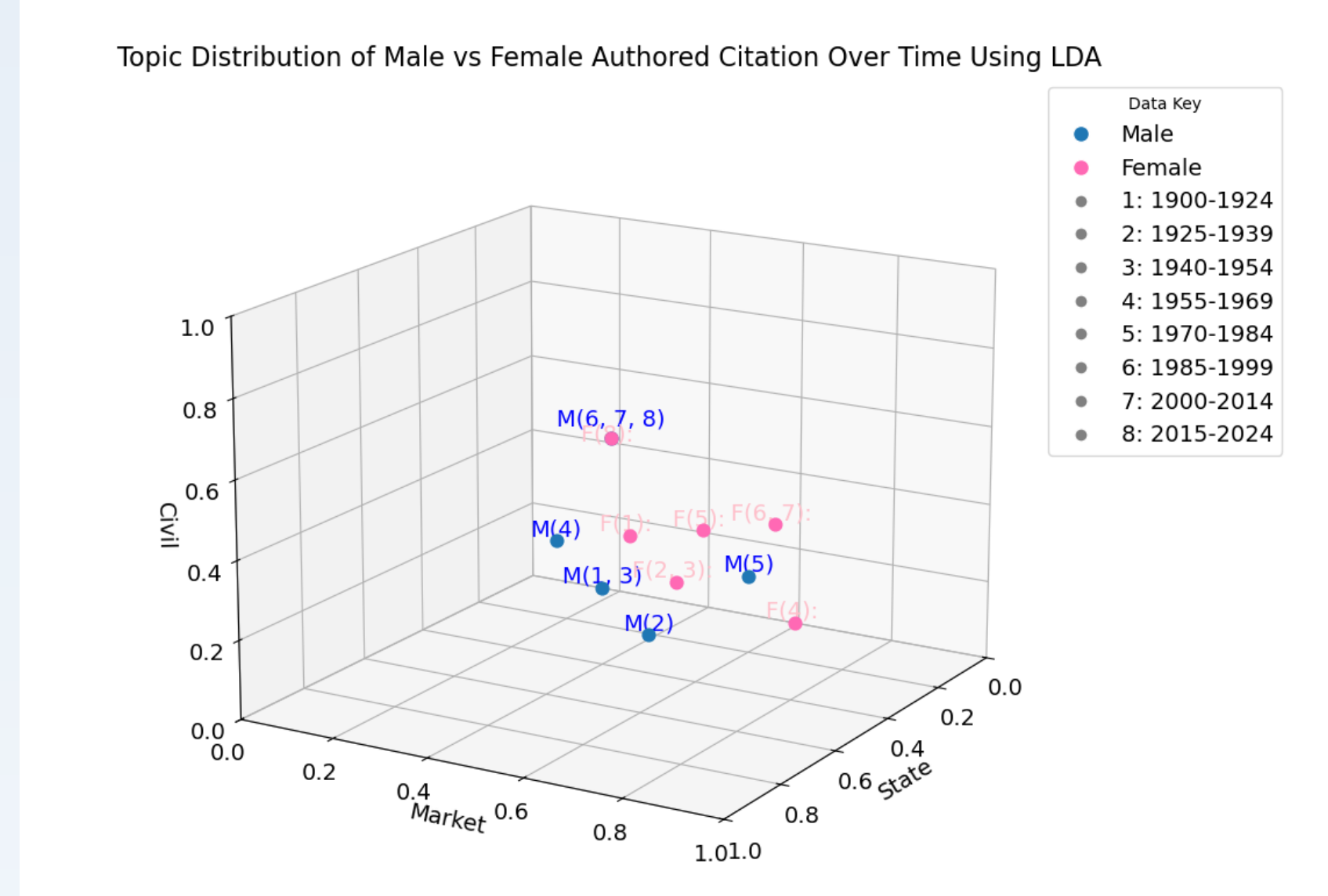
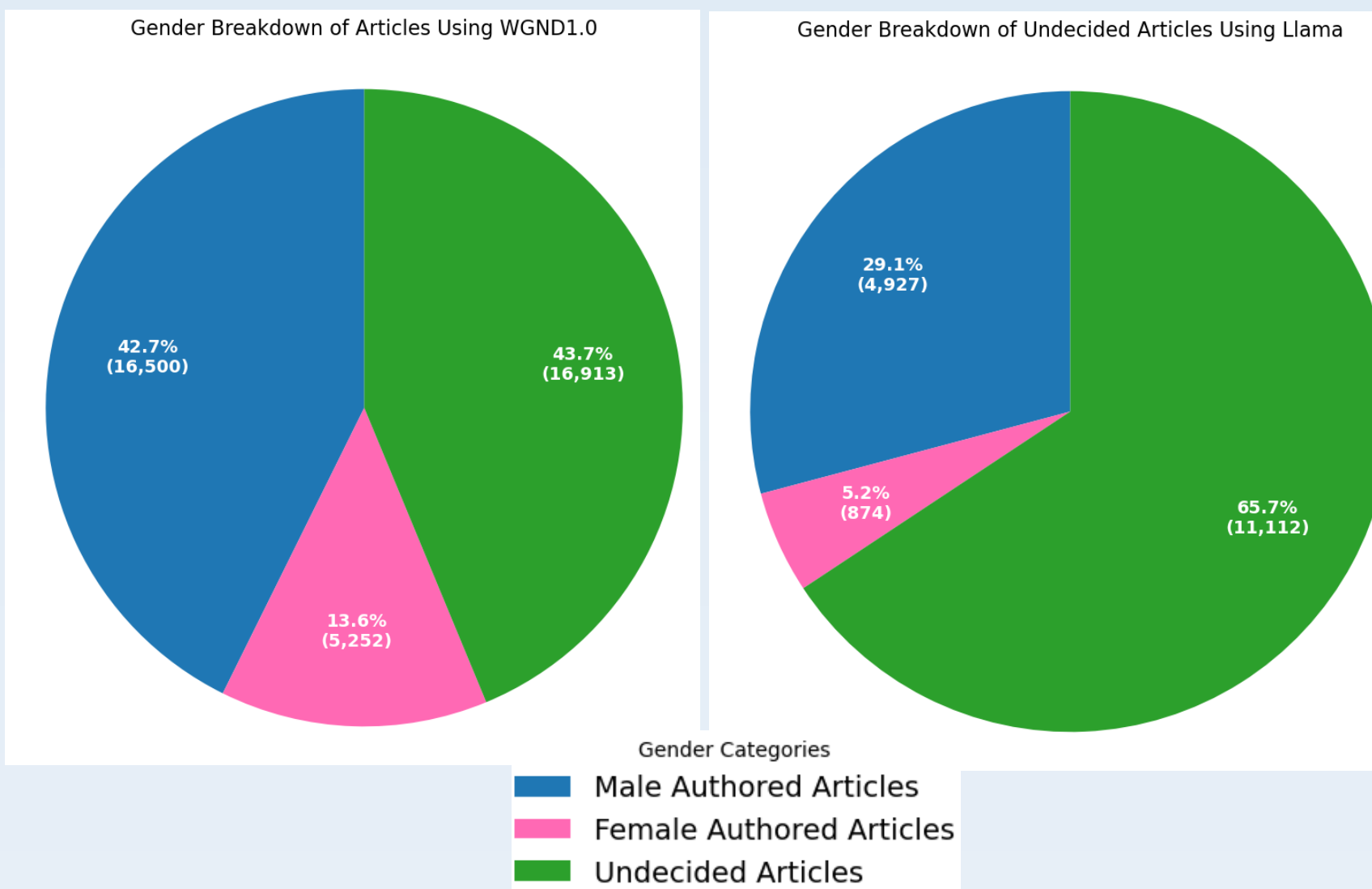
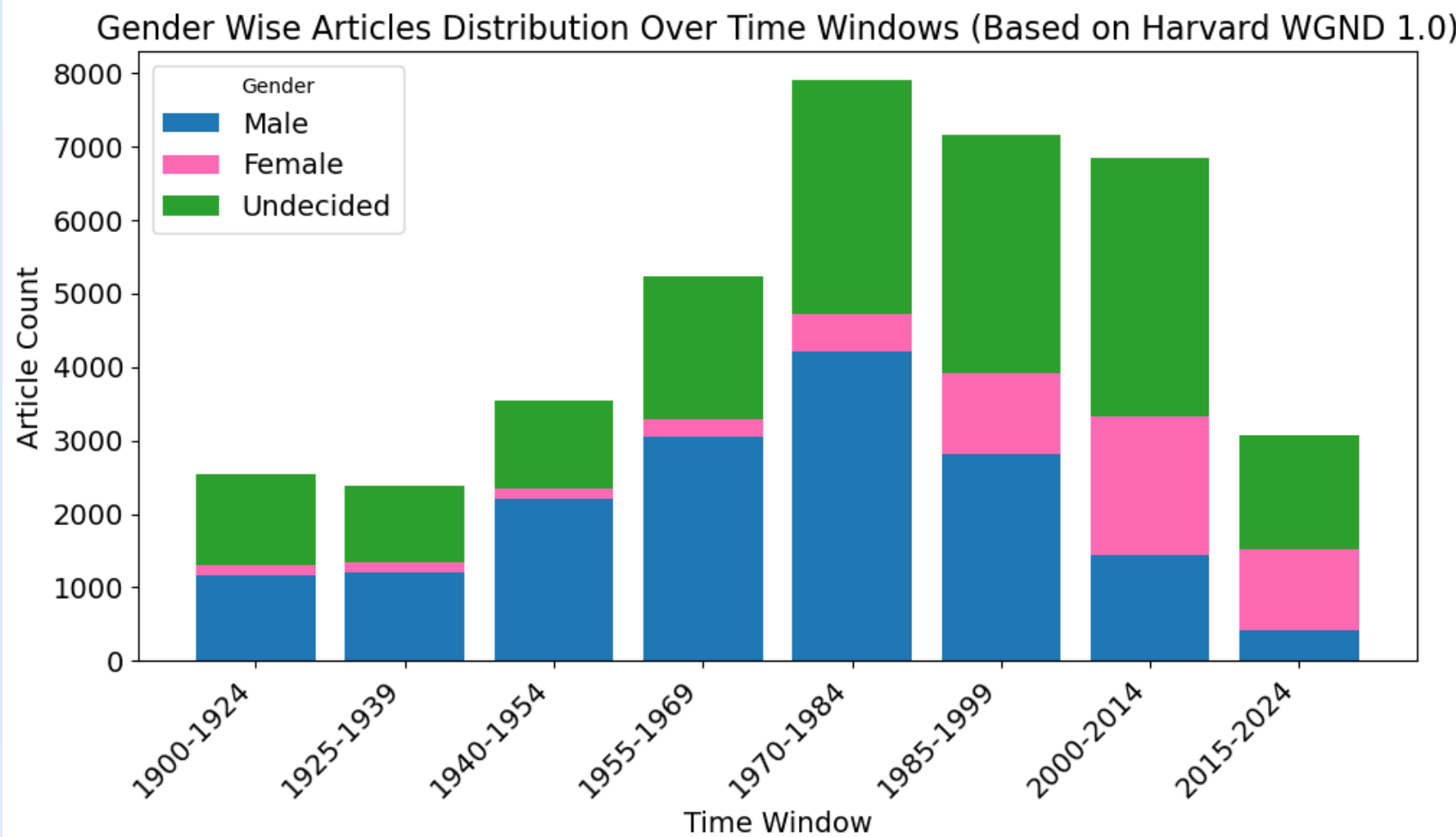
Methods



Acknowledgements

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Results



Conclusion

- Male authors dominated the early decades of economic research, but authorship increased steadily after the 1970s, particularly in market and civil society-related topics (e.g., household economics, welfare, inequality, and education).
- Female authors show stronger representation in these areas, while male authors historically focused on state and macroeconomic themes (e.g., fiscal policy, monetary policy, and trade).
- Over time, male authors' topics have diversified, increasingly engaging with civil-oriented research, though significant gender disparities in representation and thematic focus remain.
- Overall, only 13.6% of articles were female-authored compared to 42.7% male-authored, revealing a persistent gender gap in high-impact economics journals.

References

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